



Marine Envenomation/ Injury

History

- Type of bite/ sting
- Identification of organism
- Previous reaction to marine organism
- Immunocompromised
- Household pet

Signs and Symptoms

- Intense localized pain
- Increased oral secretions
- Nausea/ vomiting
- Abdominal cramping
- Allergic reaction / anaphylaxis

Differential

- Jellyfish sting
- Sea Urchin sting
- Sting ray barb
- Coral sting
- Swimmers itch
- Cone Shell sting
- Fish bite
- Lion Fish sting

Call for help/ additional resources
Stage until scene safe

Contact
Carolinas Poison Control
1-800-222-1222
Or
Agency Specific Number

	General Wound Care Procedure
	IV or IO Access Protocol UP 6 <i>if indicated</i>
P	Cardiac Monitor <i>if indicated</i>
	Drowning Protocol TE 3 <i>if indicated</i>
	Age Appropriate Allergy/ Anaphylaxis Protocol AM 1/ PM 1 <i>if indicated</i>
	Age Appropriate Hypotension/ Shock Protocol AM 5/ PM 3 <i>if indicated</i>
	Pain Control Protocol UP 11 <i>if indicated</i>

Identification of
Sea Creature

Jelly Fish
Anemone
Man-O-War

Sting Ray
Lion Fish
Urchin / Starfish

Large
Organism

Immobilize injury
Lift away tentacles Do Not brush or rub
Immerse in Hot Water 110 – 114°F (43 - 46°C) <i>if available</i>
Apply Vinegar Rinse <i>if available</i> Otherwise wash with clean seawater DO NOT use fresh water or ice

Immobilize injury
Remove Barb or Spine If large Barb in thorax or abdomen stabilize object
Immerse in Hot Water 110 – 114°F (43 - 46°C) <i>if available</i>

Immobilize injury
Multiple Trauma Protocol TB 6 <i>if indicated</i>
Extremity Trauma Protocol TB 4 <i>if indicated</i>

Monitor and Reassess

Notify Destination or
Contact Medical Control



Marine Envenomation/ Injury

Pearls

- **Ensure your safety: Avoid the organism or fragments of the organism as they may impart further sting or injury.**
- **Priority is removal of the patient from the water to prevent drowning.**
- **Coral:**
 - Coral is covered by various living organisms which are easily dislodged from the structure.
 - Victim may swim into coral causing small cuts and abrasions and the coral may enter into cuts, causing little if any symptoms initially, but later causing inflammation, pain and/ or infection.
 - The next 24 – 48 hours may reveal an inflammatory reaction with swelling, redness, itching, tenderness, and ulceration.
 - Treatment is flushing with large amounts of fresh water or soapy water then repeating.
- **Jelly Fish/ Anemone/ Man-O-War:**
 - Wash the area with fresh seawater to remove tentacles and nematocysts.
 - Do not apply fresh water or ice as this will cause nematocysts firing as well.
 - Recent evidence does not demonstrate a clear choice of any solution that neutralizes nematocysts.
 - Vinegar (immersion for 30 seconds), 50:50 mixture of Baking Soda and Seawater, and even meat tenderizer may have similar effects.
 - Immersion in warm water for 20 minutes, 110 – 114°F (43 - 46°C), is effective in pain control.
 - Shaving cream may be useful in removing the tentacles and nematocysts with a sharp edge (card).
 - Stimulation of the nematocysts by pressure or rubbing cause the nematocyst to fire even if detached from the jellyfish.
 - Lift away tentacles as scrapping or rubbing will cause nematocysts firing.
 - Typically symptoms are immediate stinging sensation on contact, intensity increases over 10 minutes.
 - Redness and itching usually occur.
 - Papules, vesicles and pustules may be noted and ulcers may form on the skin.
 - Increased oral secretions and gastrointestinal cramping, nausea, pain, or vomiting may occur.
 - Muscle spasm, respiratory, and cardiovascular collapse may follow.
- **Lionfish:**
 - In North Carolina this would typically occur in a residence/ business as lionfish are often kept as pets in saltwater aquariums.
 - Remove any obvious protruding spines and irrigate area with copious amounts of saline.
 - The venom is heat labile so immersion in hot water, 110 – 114°F for 30 to 90 minutes is the treatment of choice but do not delay transport if indicated.
- **Stingrays:**
 - Typical injury is swimmer stepping on ray and muscular tail drives 1 – 4 barbs into victim.
 - Venom released when barb is broken.
 - Typical symptoms are immediate pain which increases over 1 – 2 hours.
 - Bleeding may be profuse due to deep puncture wound.
 - Nausea, vomiting, diarrhea, muscle cramping, and increased urination and salivation may occur.
 - Seizures, hypotension, and respiratory or cardiovascular collapse may occur.
 - Irrigate wound with saline. Extract the spine or barb unless in the abdomen or thorax, Contact Medical Control for advice.
 - Immersion in hot water, if available, for 30 to 90 minutes but do not delay transport.
- Patients can suffer cardiovascular collapse from both the venom and/ or anaphylaxis even in seemingly minor envenomation.
- Sea creature stings and bites impart moderate to severe pain.
- Arrest the envenomation by inactivation of the venom as appropriate.
- Ensure good wound care, immobilization and pain control.